

VERITA

TECHNICAL APPENDIX

the system, the model, and its honest limits

```
verita@review  
  
$ git log --oneline  
* interview-ready  
* tests green  
* chain verified  
$ ready _
```

0.913

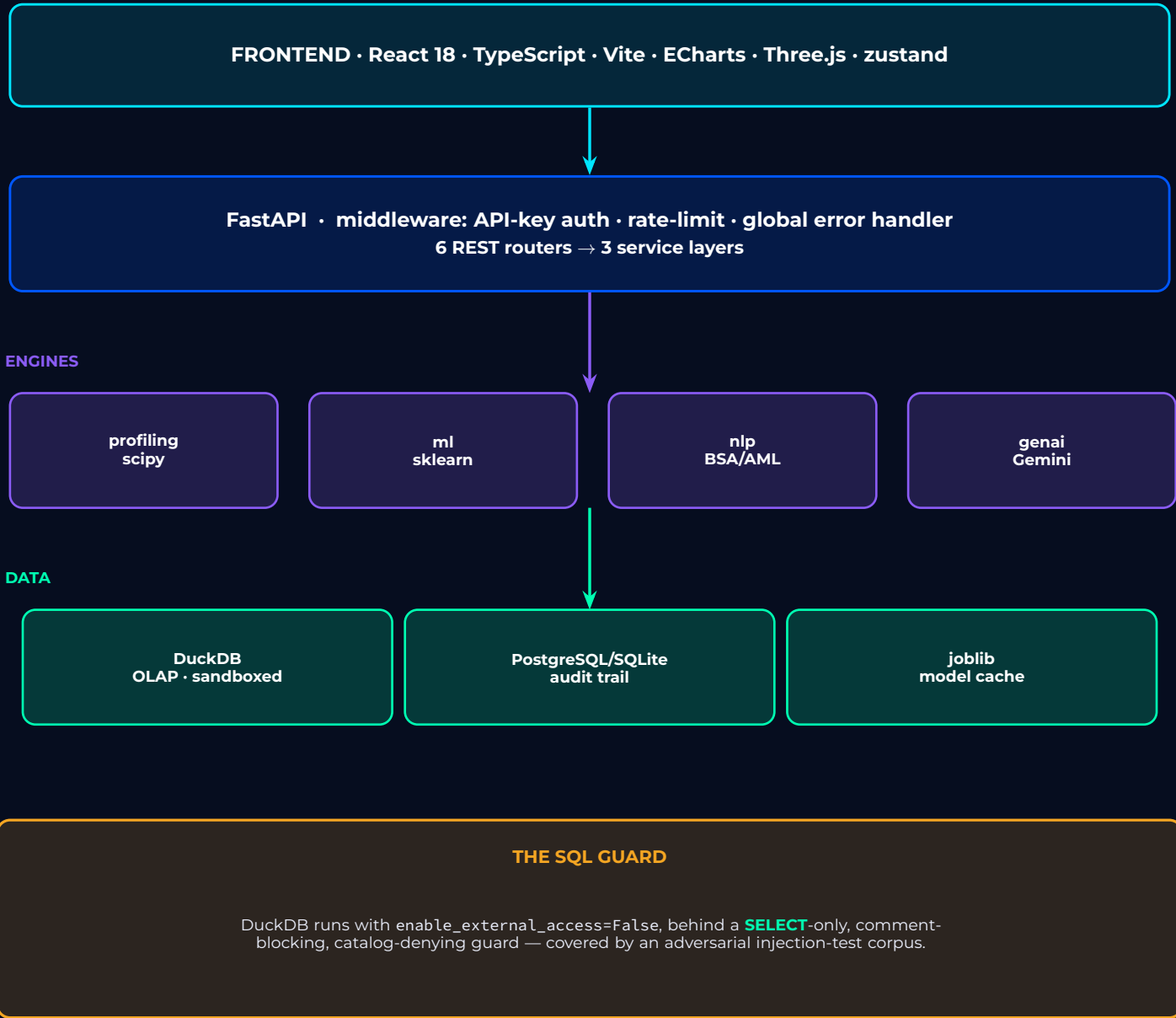
ROC-AUC

0.65

PR-AUC

284K

real tx



Engineering posture: disk-backed store (uploads survive restart) · model persisted (0.36s boot vs 21s retrain) · profiles cached once.



CLASS IMBALANCE · why this is hard

284,807 transactions

0.17% fraud ← the red sliver

MODEL CONFIG

```
GradientBoosting
n_estimators=120
max_depth=3
learning_rate=0.12
subsample=0.85
IsolationForest
contamination=rate
```

WHY IT'S FAST

Trains once (~2 min), then the fitted engine is **persisted to joblib**. Every later boot **loads in 0.36 s** instead of retraining — the model is a cached artifact, not a request-time cost.

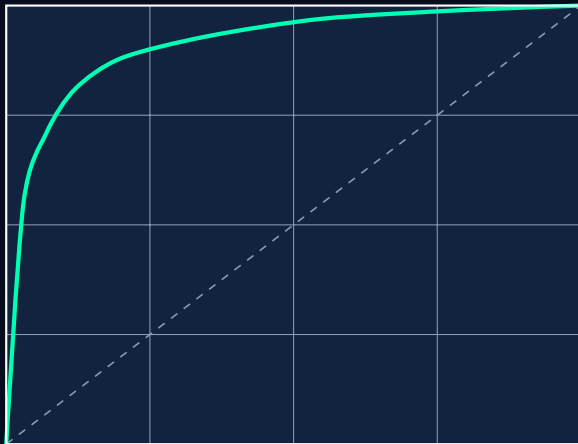
💡 WHY GBM, NOT DEEP LEARNING?

Tabular data + 0.17% positives → gradient-boosted trees win, dependency-free. XGBoost + Optuna is the roadmap, not a need for this dataset.

THE UNGLAMOROUS PARTS, DONE RIGHT

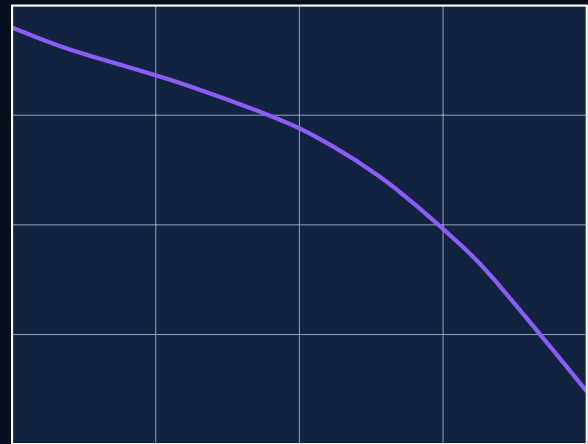
Real fraud data (not synthetic), a stratified split (not a lucky one), a persisted model (not a 21-second retrain on every boot). Boring on purpose — that is what production looks like.

ROC CURVE · AUC 0.913



false positive rate →

PR CURVE · AP 0.65



recall → (precision ↑)

CONFUSION MATRIX · structure

	pred —	pred +
true —	TN	FP
true +	FN	TP

$$\text{precision} = \frac{TP}{TP + FP}$$

$$\text{recall} = \frac{TP}{TP + FN}$$

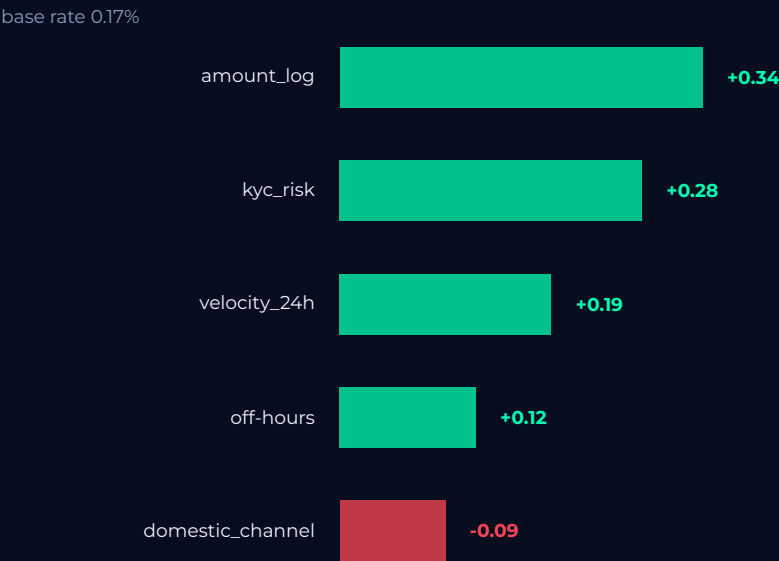
Exact cell counts move with the decision threshold — the live slider shows the precision/recall trade-off in real time.

💡 WHY PR-AUC IS THE HEADLINE

With positives at 0.17%, ROC-AUC flatters the model. PR-AUC reflects the precision/recall trade-off the AML alert queue actually operates on. **5-fold stratified CV** confirms the held-out score isn't a lucky split, and **precision@k** is what the analyst queue is judged on.

the numbers you see are the numbers a regulator gets — measured, not chosen.

WATERFALL · one prediction's anatomy



f(x) = 0.79 risk score

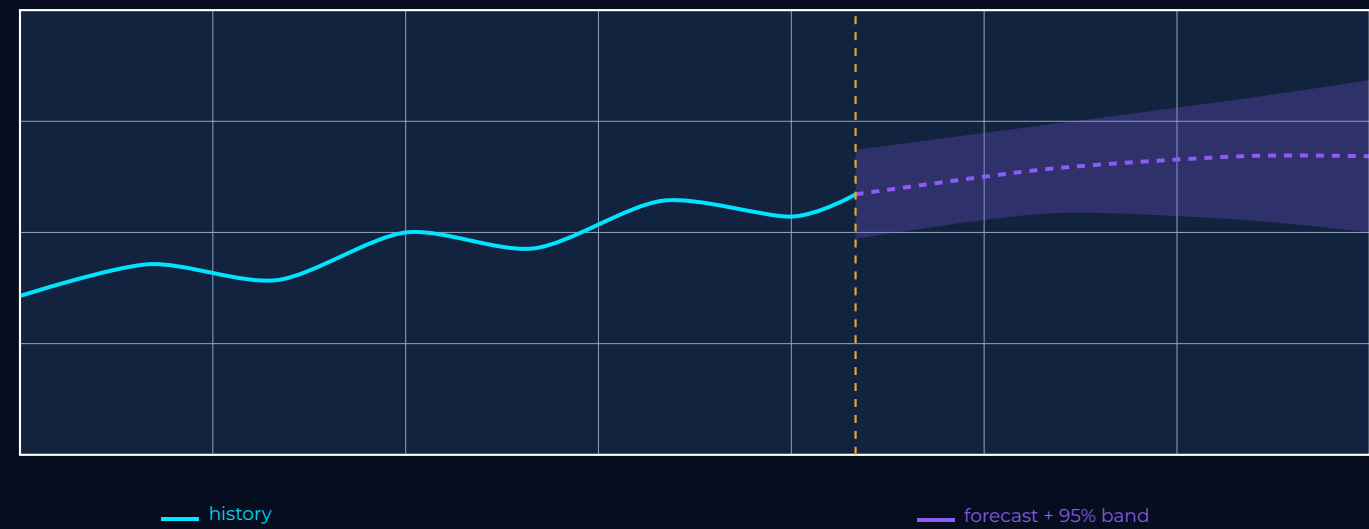
 ANALYST-READABLE EXPLANATION

"Flagged mainly because **amount_log** (+0.34), **kyc_risk** (+0.28) and **velocity** (+0.19) pushed risk up; the domestic channel (−0.09) pulled it down. Net score **0.79**." — generated server-side from the SHAP values, deterministic.

 WHAT THIS MODEL CANNOT CLAIM

Trained on *one issuer's* 2013 EU card data; V1, .V28 are PCA-anonymised (no human-readable semantics). Not validated on wire/ACH/crypto rails; no concept-drift monitoring yet; not a substitute for a tuned production sanctions screen.

FORECAST · history + projection + 95% band



4-MODEL TOURNAMENT · held-out MAPE (lower = better)



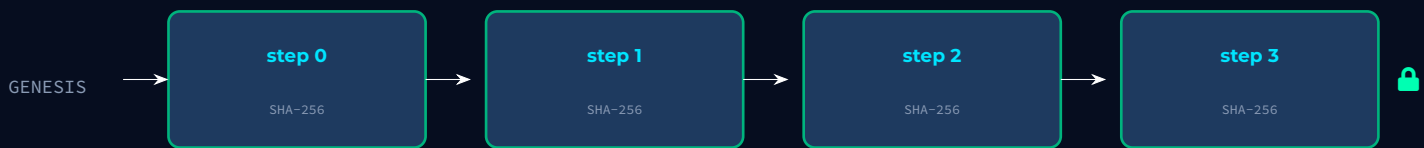
WINNER RATIONALE

Holt-Winters wins because it captures the trend the naïve baseline misses — and it must *beat* the naïve benchmark to be chosen. The band is $\pm 1.96\sigma$ of holdout residuals, not a made-up $\pm 15\%$.

REPORTED PER MODEL

MAPE · SMAPE · MASE — plus the margin over the naïve baseline, scored with **rolling-window** backtesting, not a single split.

THE HASH CHAIN · tamper-evident trace



each step folds in the previous hash and commits `id · query · finding · severity · timestamp · prev-hash`

5 REAL AML TYPOLOGIES, each tested with a DuckDB query + statistic

- ✓ Geographic risk concentration
- ✓ Channel risk concentration
- ✓ Structuring near the \$10,000 threshold
- ✓ Large-value concentration (top-1%)
- ✓ Temporal spike (mean + 2σ)

compliance-memo.txt

```
COMPLIANCE INVESTIGATION MEMO – Overall risk: HIGH
1. [High] Jurisdiction X: 3.1x baseline fraud rate (n=412)
2. [Critical] 47 tx cluster just below $10,000 (structuring)
Recommended action: open EDD; prepare SAR within 5 days.
> every line backed by a logged, reproducible query.
```

doctor any step → the chain breaks.

REPRODUCIBLE, WITH OR WITHOUT THE LLM

The reasoning is deterministic — the engines decide the findings; an LLM only narrates the memo when a key is present. A regulator can re-run the whole investigation offline.

Pandas sync profiling

Polars/Dask or chunked streaming; async job pipeline first.

In-memory rate limiter

Redis token-bucket shared across workers.

Single-node DuckDB

MotherDuck / external OLAP for multi-GB data.

No real OFAC integration

Wire the SDN list + fuzzy name matching into the NLP matcher.

No human feedback loop

Capture analyst TP/FP disposition → retraining set.

No drift monitoring

PSI/KS on feature & score distributions; alert on drift.

It works.

It's documented.

It was built for Wolters Kluwer.

> ask me anything.